



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECD263 - SYSTEM ANALYSIS AND DESIGN (SEC 01)
SEMESTER II (2025/2026)

Project Phase1:

GROUP: ONGMARI

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1.0 Introduction

The FlexHub Artificial Intelligence (AI) Marketplace includes FlexRent, which is an intelligent platform that handles property rentals with increased scheduling flexibility. In today's market, the rental management on current platforms is restricted by static models. This leads to poor user experience as the requirements from both "Time" and "Style" are not met. Therefore, users are often disappointed by the limited choices and forced to look for better platforms.

The FlexRent platform could solve the challenges faced by users because it uses a specially designed system that integrates AI technologies such as Smart Matching, Smart Filter, Smart Scheduler and Dynamic Pricing. These functions not only offer personalized recommendations to users but also block overlapping time slots and encourage early booking. Smart filters also allowed the users to get the venue that they were interested in rather than finding the venue by scrolling the web page for multiple hours. Besides that, time-slot management is applied to control conflicts risk detection and allocation for buffer times. This feature allows the staff to estimate total time acquired by users with improved accuracy. Moreover, dual role management ensures the change of users between clients and owners. This is highly convenient that it could attract users who want to play both roles as only one account is used and users can simply switch between "Seekers Mode" and "Providers Mode". Hence, this system is primarily designed for users and it will benefit the staff as well.

2.0 Background Study

Nowadays, online platforms have changed the way people rent properties and spaces. Many users prefer to book accommodations or venues through digital platforms because it is faster and convenient. For example, platforms like Airbnb focus on short-term stays, while other platforms such as PropertyGuru are more suitable for long-term rentals.

However, most of these systems are designed for specific types of rentals only. Some platforms only allow daily bookings, while others focus on monthly rentals. There are also systems for event spaces that support hourly booking, but they are usually separate from other platforms. Thus, users often need to switch between different systems depending on their needs, which can be inconvenient.

In addition, many existing systems do not include advanced features such as intelligent recommendations or smart scheduling. Users usually need to manually search and compare options, which can take time. The system also does not always consider issues like overlapping bookings or the need for buffer time between reservations.

Therefore, there is a clear gap in the current market for a more flexible and intelligent system that can support different rental durations (hourly, daily and monthly) in one platform, while also improving the overall user experience.

3.0 Problem Statement

Even though there are many online rental platforms available today, several problems still exist. Firstly, most systems do not provide flexible rental options. They usually focus on only one type of booking, such as daily or monthly. This becomes a problem when users need different rental durations, as they are forced to use multiple platforms instead of one complete system.

Besides that, booking conflicts and inefficient time management are still a common issue. Current systems do not effectively handle overlapping bookings and incorporate buffer time between reservations for activities such as cleaning or preparation. This may lead to double bookings, service delays and poor user satisfaction.

Apart from that, the lack of smart recommendation features makes it harder for users to find suitable properties. Most systems only provide basic filtering rather than personalized suggestions. Therefore, users have to spend more time searching for options that match their needs, budget and preferred time without AI-driven matching.

Additionally, the lack of integrated scheduling and decision support reduces system efficiency. Property owners still need to manage bookings manually in many cases. This increases the chances of human error and makes the process less efficient.

Overall, these issues show that current rental platforms are not flexible or intelligent enough, and highlight the need for a more advanced and flexible rental platform that leverages artificial intelligence to improve scheduling, recommendations and system efficiency.

4.0 Proposed Solutions

Our challenge is that the existing platforms now are too “static.” FlexHub aims to be an intelligent system that understands that “Time” and “Style” are different for every user. Whether someone is renting a wedding hall for 5 hours or an apartment for 12 months, the experience should be seamless. Therefore, our team plans to have AI integration, time-slot management and role management as our core requirements in the platform.

First and foremost, the platform will use AI integration to help the user find a suitable place they want to rent. The system includes both AI recommendation and smart filtering features. The AI recommendation function suggests venues based on the user's preferences such as budget, purpose and preferred time, while the smart filtering feature allows the user to narrow down options by location, capacity, type and price. Unlike basic filters, it can also consider user preferences to provide more relevant results, and users can get a better experience in selecting the venue they favour.

Furthermore, the platform has time slot management. This management enables the smooth connection between the previous customer and the upcoming customer to use the venue. In other words, this management will handle conflict and “buffer times” between bookings. The “buffer times” allow the owner of the venue to clean up the place before the next customer comes. This ensures the cleanliness of the venue and maintains the customer's rating of the venue rent.

Last but not least, the role management in the platform allows the system to distinguish among the venue owner, admin and customers. The users can be both “Seekers” (Renters/Clients) and “Providers” (Owners/Experts). This role management helps to ease the work of users. They will enter a different page according to their identity. Admin and staff can review the data in the platform

and analyze the data for each month. Staff will also monitor the payment process to ensure the safety of transactions.

4.1 Feasibility Study

A feasibility study is a detailed evaluation carried out before beginning a project, business, or system to ascertain whether it is technically possible, financially feasible, and practically achievable. Its key purpose is to help the project manager understand whether the idea should be developed and modified before investing a substantial amount of time, funds and resources. In a feasibility study, three distinct aspects of the proposed project are analysed systematically.

4.1.1 Technical Feasibility

Technical feasibility assesses whether the proposed technical resources are able to be implemented for the new system.

This rental system will be designed as a web-based platform using modern technologies such as HTML, C++, JavaScript and a backend framework. The project will be carried out using standard computers and widely available software tools, which make it affordable and accessible. The development team requires the necessary programming technique and system design skills to implement the platform successfully and effectively. The system includes standard features such as a user menu, booking management and payment processing, which are achievable with current technology. Additionally, the platform is made to be scalable and maintainable, which allows for future enhancement and updates.

4.1.2 Operational Feasibility

Operational feasibility determines if the human resources are available to operate the system once it has been installed in order to meet the needs of users.

The rental platform is designed with a simple and user-friendly interface. This makes it much easier for renters and venue owners to navigate and use without requiring technical knowledge. The rental platform provides direct features such as smart searching, reservation and online payment to ensure users can make their booking efficiently. This can replace manual booking processes with an automated smart platform and improve efficiency by reducing human errors. Moving on, a comprehensive training process is provided to the new staff and basic user guidelines will be provided for every new user to understand the features inside the platform.

4.1.3 Economic Feasibility

In economic feasibility, it determines whether the time and money are available to develop the rental system by performing a cost-benefit analysis (CBA) to evaluate the budget. The budget includes the purchase of new equipment. For instance, hardware and software. Cost-Benefit Analysis (CBA) analysis aims to compare costs and benefits to see whether investing in the development of a new system will be beneficial. CBA techniques consist of break-even analysis, payback, cash-flow analysis and present value analysis. There are tangible benefits and intangible benefits. For the expenses, there are development costs and production (operation) costs. By knowing the exact budget, the project can be managed more smoothly.

5.0 Objectives

The objectives of the proposed FlexRent System are:

1. To develop a flexible web-based rental platform that supports multiple rental durations such as hourly, daily and monthly bookings in a single system. This helps users avoid switching between different platforms for different rental needs.
2. To implement intelligent recommendation features that can suggest suitable venues based on users' preferences, budget, purpose and preferred time. This improves user experience and reduces the time needed to search for appropriate rental options.
3. To design an effective booking and time-slot management system that can detect overlapping bookings, manage buffer time and prevent double booking. This ensures smoother scheduling and better operational efficiency for both users and venue owners.
4. To provide a secure and convenient online transaction process by integrating payment methods such as online banking, debit/credit card and e-wallet. This ensures that booking transactions are safe, fast and reliable.
5. To improve overall system efficiency for users, venue owners and administrators by automating rental processes such as booking, confirmation, notifications and record management. This reduces manual work, minimizes human error and enhances service quality.

6.0 Scope of the Project

1. User Login and Sign-Up Menu

- New users can register an account here.
- Users can choose their role as customer, provider or admin.
- Allow users to log in and edit their profile.
- Users need to enter their username or email and password to log in.
- If a user forgets their password, they can request a temporary password and we will email it to them.

2. Venue Listing Site

- The owner can upload their venue to rent, including an outdoor space, a meeting hall, a house or even a studio.
- The owner can upload the venue photo and the facilities available.
- The owner can fill in the details of their venue, such as address, price, date and time slot.
- The owner can hide the venue if it is not available.

3. Smart Searching and Filter Feature

- Users can search for the type of venue they require.
- This feature can also filter out the venue that customer requests.
- This filter feature categorizes the venue based on the location, capacity, type and price.
- From this feature, users can see the overall view of the venue that the platform has to rent.
- This feature helps users to save time in scrolling to search for a suitable venue they need by considering the venue size and even price as well.

4. Venue Detail Page

- On this page, users can go through the details of the venue selected.
- There is a description for the venue along with the pictures of what the venue looks like.
- If the customer is satisfied with the venue, they can choose the time slot by referring to the availability of the venue.
- The owner's number is also available on this site for the customer to do further surveys for the venue.

5. Booking System

- Core feature for users to reserve a venue on a specific date and time.
- The system will check the availability of the venue before the booking request is approved.
- This system helps to prevent double booking.
- Once confirmed, the booking details will be stored in the system and shared with both the user and venue owner.
- The venue selection will also be closed for 10 minutes for the payment process.

6. Online Payment System

- Users only have 10 minutes to complete the transaction process.
- It ensures that transactions between users and venue owners are safe, fast and reliable.
- A variety of payment methods can be selected to ease our users to complete the payment.
- Payment methods include online banking, debit/credit card, or e-wallet.

7. Booking Confirmation and Notifications

- After payment is done, users will receive a digital receipt and booking confirmation notice.
- The platform will also notify the user and venue's owner before the day they rent the place.
- This notification prevents the user from forgetting the time.
- Notifications also remind the venue's owner to clean up the venue and ensure the venue is clean and tidy before the customer comes.

8. User Reviews and Ratings Site

- After the user experiences the venue they rent, they can give feedback and comment their feelings.
- Users can rate their satisfaction through the platform.
- Admin and owner of the venue can view the feedback and make further improvements if needed.

9. Admin Panel

- Admin needs to log in by using their staff ID and password.
- Admin has full access to view the data in the platform.
- Staff have half access to view the information.
- Admin manages users and venue listings.
- Staff can access and analyze the data in each system.
- Staff can improve user experience and update the system to fulfill the user's needs.
- Staff will also assist users if they encounter any problems.

7.0 Project Planning

7.1 Human Resource

Name	Role	Key Responsibilities
Ling Xiao Xuan	Project Manager	- Plan the project by setting the goals, timeline and budget - Assign tasks to team members to ensure smooth flow in the project - Monitor progress and make sure work is on schedule and done before the due date - Communicate with team members to prevent misunderstanding - Overcome the problems and manage risks throughout the project
Lok Jie Ying	System Developer	- Discover various features such as user registration, venue listings, search filters, booking system and payment integration - Connect the frontend (what users see) with the backend (database, server) - Write code for the features and fix bugs - Maintain and update the system after launch from time to time

		- Ensure the system is efficient and meets user needs and style
Emily Tai Pei En	UI/UX Designer	- Design user-friendly screens such as the search page, booking page and payment page - Ensure it is convenient for users to browse and book venues - Improve user experience - Design the layout of the platform (homepage, search page, booking page) - Improve the system based on user experience and their feedback
Ma LiWei	System Analyst	- Identify features such as search venues, booking system, payment and reviews - Gather requirements from stakeholders (users, venue owners) - Communicate between the developer and stakeholders - Design system flow diagrams or models - Translate project needs into technical requirements for developers

7.2 Work Breakdown Structure (WBS) - Product-Based WBS

Level 1: Web-Based FlexRent System

- Level 2: Major System Components

1. Frontend System

1.1 User Registration Page (login and sign up)

1.2 User Profile Information Page (name, age, gender and contact details)

1.3 User Satisfaction Section (comments and ratings)

1.4 Venue Dashboard

1.5 Calendar Section

1.6 Venue Booking Confirmation Page

1.7 Venue Check-in & Check-out Page

1.8 Online Transaction Interface (card, e-wallet, bank transfer)

1.9 User Preference Page (length of time, size of venue, purpose)

2. Role Management Module

2.1 “Seekers” Selection (rental or clients)

2.2 “Provider” Selection (owners or experts)

3. Backend System

3.1 User Verification System (through email or contact number)

3.2 Admin Panel

3.3 Rent Order and Sales Management

3.4 Inventory Reports Module

4. Artificial Intelligence (AI) Integration Feature

4.1 Smart Matching (according user's needs)

4.2 Smart Scheduler (based of preferred time-slot)

4.3 Dynamic Pricing (for holidays and weekends)

5. Time-Slot Management System

5.1 Venue Information & Status Module

5.2 Buffer Time Control (clean, travel, inspection)

5.3 Conflict Handling

5.4 Unit of Measure (hour, day, month)

5.5 Itinerary Generation

6. Payment and Security

6.1 Fraud Risk Detection

6.2 Data Privacy Protection

6.3 Admin Access Authorization

6.4 Invoice & Receipt Generation

6.5 Pricing Calculation (hour, day, month)

7. Testing and Quality Assurance

7.1 AI Performance Logic Testing

7.2 Security Assurance

7.3 Rental Unit Testing

7.4 User Acceptance Testing

7.5 Bug Retesting

8. Deployment and Maintenance

8.1 Hardware Maintenance

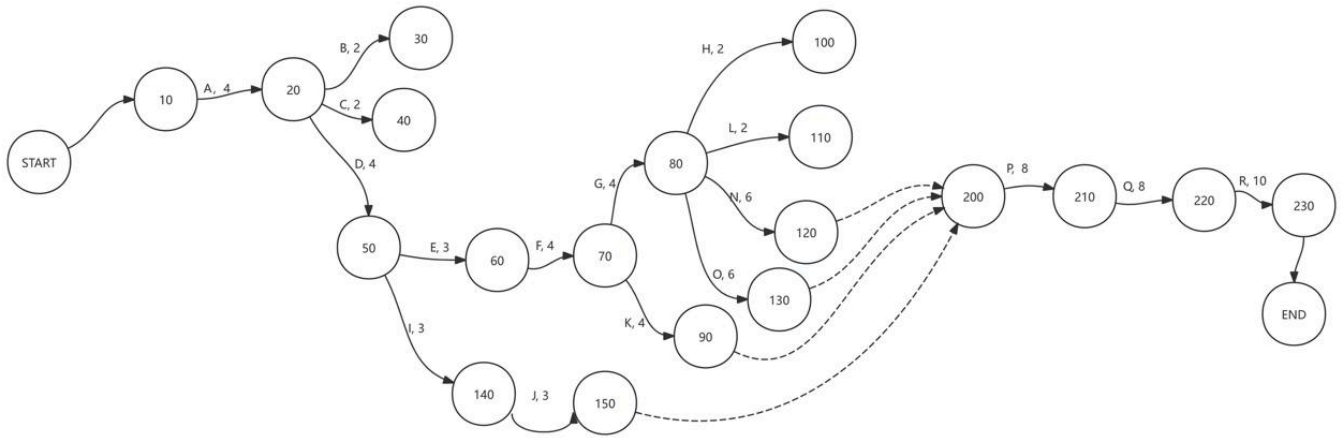
8.2 Software Updates

8.3 Staff & User Training

8.4 Feasibilities Evaluation

8.5 AI Training

7.3 PERT Chart (based on WBS)

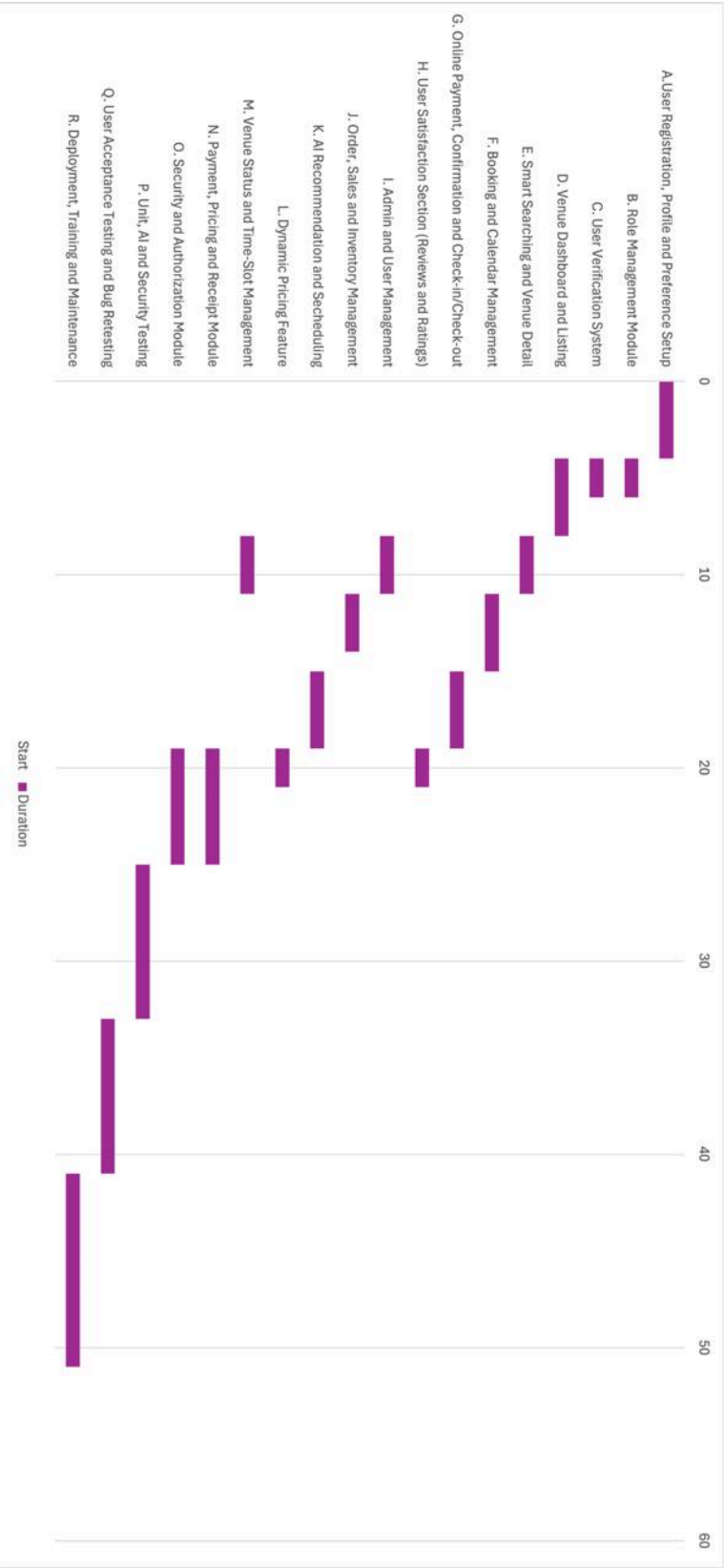


Letter	Activity	Predecessors	Duration(Days)
A	User Registration, Profile and Preference Setup	-	4
B	Role Management Module	A	2
C	User Verification System	A	2
D	Venue Dashboard and Listing	A	4
E	Smart Searching and Venue Detail	D	3

F	Booking and Calendar Management	E	4
G	Online Payment, Confirmation and Check-in/Check-out	G	4
H	User Satisfaction Section (Reviews and Ratings)	G	2
I	Admin and User Management	D	3
J	Order, Sales and Inventory Management	I	3
K	AI Recommendation and Scheduling	A,F	4
L	Dynamic Pricing Feature	G	2
M	Venue Status and Time-Slot Management	D	3

N	Payment, Pricing and Receipt Module	G	6
O	Security and Authorization Module	G	6
P	Unit, AI and Security Testing	J, K, M, N, O	8
Q	User Acceptance Testing and Bug Retesting	P	8
R	Deployment, Training and Maintenance	Q	10

7.4 Gantt Chart



8.0 Benefit and Overall Summary of Proposed System

The FlexRent System should be implemented in the hospitality field as it has diverse benefits for the staff and customers, ensuring satisfaction from both staff and customers. One primary benefit is application of Artificial Intelligence (AI) technologies, Smart Filters, Smart Matching, Smart Scheduling and Dynamic Pricing. These functions allow users to find their desired venue quickly. For instance, Smart Matching and Smart Filters automatically suggests the related venue after the users enters their preference during the registration interface and the less preferred will seldom appear on the homepage. This results in a smoother booking experience for customers and enhances customer acceptance towards the system, raising the profile and business value of the accommodations.

Furthermore, the workload of staff is reduced and a more effective workflow is created. Miscommunication due to unclear handwriting or misinterpretation of requirements from customers could be prevented as the bookings and customer details are stored in a database in the computers. The data are always accessible through a stable Internet connection. In this way, staff performance is improved and more time is saved to complete other tasks such as buffer time adjustment and sudden change of booking cancellation. The problem of overworking is resolved, encouraging better well being of staff and a positive working environment.

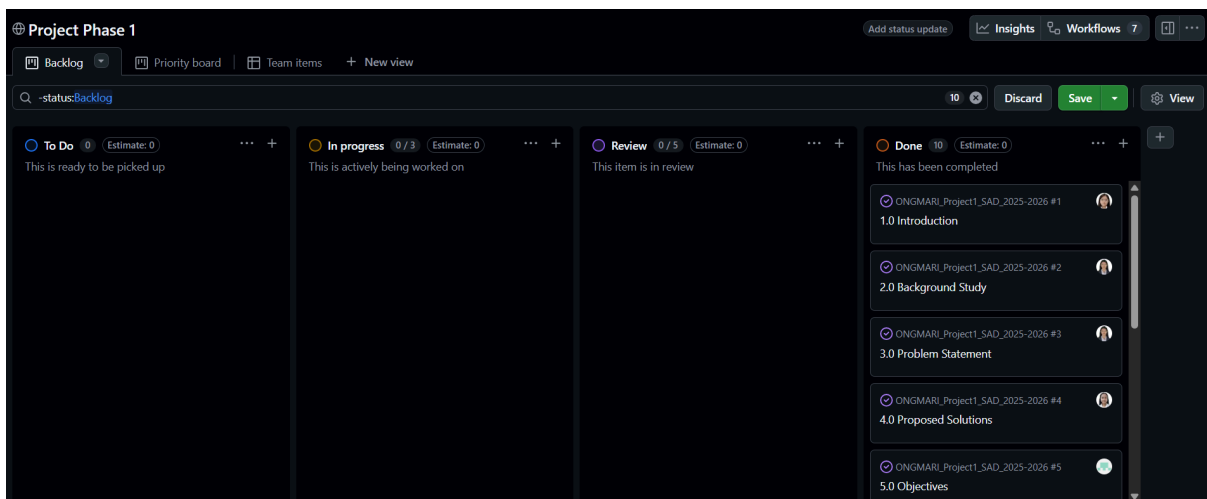
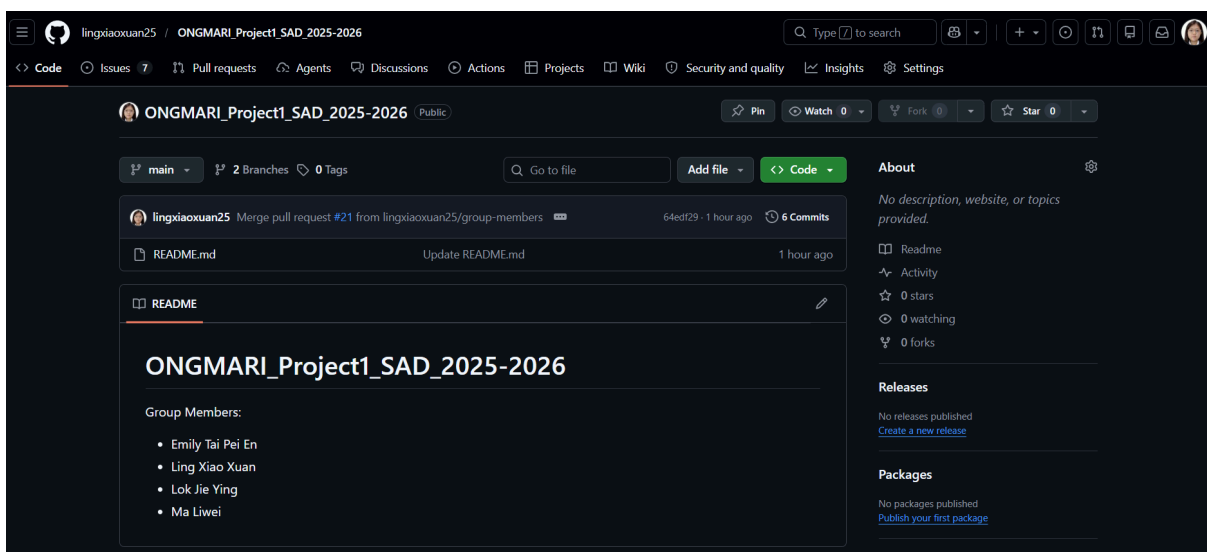
Moreover, the platform would recommend new system installation and notify hardware compatibility for long term technical support. This is crucial so that the proposed system could operate at all times and the booking process is not affected, fostering a bond between staff and users . For example, hardware such as computers and printers should be regularly maintained and the latest version of software is supported. The safety of customer data is guaranteed through data privacy compliance as well, building trust between the system and users.

To sum up, the system is a powerful tool which comprises a wide range of functions, allowing users to make bookings without facing obstacles. This platform is dynamic towards the demand of users and should fully replace the manual, time-consuming processes. As technological innovations bring advancement to human life, this platform should be expanded across cities, countries and worldwide.

9.0 Project Management

Github Repository Link:

https://github.com/lingxiaoxuan25/ONGMARI_Project1_SAD_2025-2026.git



Project Phase 1

Backlog

Priority board

Team items

New view

Assignees

Deselect

EmilyEnnn3

lingxiaoxuan253

lokjeying2

maliwei9072

Show empty values

Q

-status:Backlog

3

Discard

Title

...

Status

...

Done3

Estimate: 0

This has been completed

...

1

4.0 Proposed Solutions #4

Done

...

2

6.0 Scope of Project #6

Done

...

3

7.1 Human Resource #7

Done

...

+

Add item

Project Phase 1

Backlog

Priority board

Team items

New view

Assignees

Deselect

EmilyEnnn3

lingxiaoxuan253

lokjeying2

maliwei9072

Show empty values

Q

-status:Backlog

3

Discard

Title

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Status

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Done3

Estimate: 0

This has been completed

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1

1.0 Introduction #1

Done

...

2

7.2 Work Breakdown Structure (WBS) #8

Done

...

3

8.0 Benefit and Overall Summary of Proposed System #10

Done

...

+

Add item

Project Phase 1

Backlog

Priority board

Team items

New view

Assignees

Deselect

EmilyEnnn3

lingxiaoxuan253

lokjeying2

maliwei9072

Show empty values

Q

-status:Backlog

2

Discard

Title

...

Status

...

Done2

Estimate: 0

This has been completed

...

1

2.0 Background Study #2

Done

...

2

3.0 Problem Statement #3

Done

...

+

Add item

Project Phase 1

Backlog

Priority board

Team items

New view

Assignees

Deselect

EmilyEnnn3

lingxiaoxuan253

lokjeying2

maliwei9072

Show empty values

Q

-status:Backlog

2

Discard

Title

...

Status

...

Done2

Estimate: 0

This has been completed

...

1

5.0 Objectives #5

Done

...

2

7.3 PERT Chart & 7.4 Gantt Chart #9

Done

...

+

Add item

25

Version Control Best Practices	Importance
Feature branching	- Allows collaboration between group members as they can work simultaneously - Avoids conflicts as merges are controlled - The progress is shared without affecting collaborators, providing parallel development
Pull requests	- Enables PR creation and feedback handling - Changes made have detailed records of discussion and approval - Increase work visibility
Code reviews	- Prevent bugs issues - Enhance consistency as the standard on patterns and style are agreed by all members - Receive useful insights from mentors to improve written code